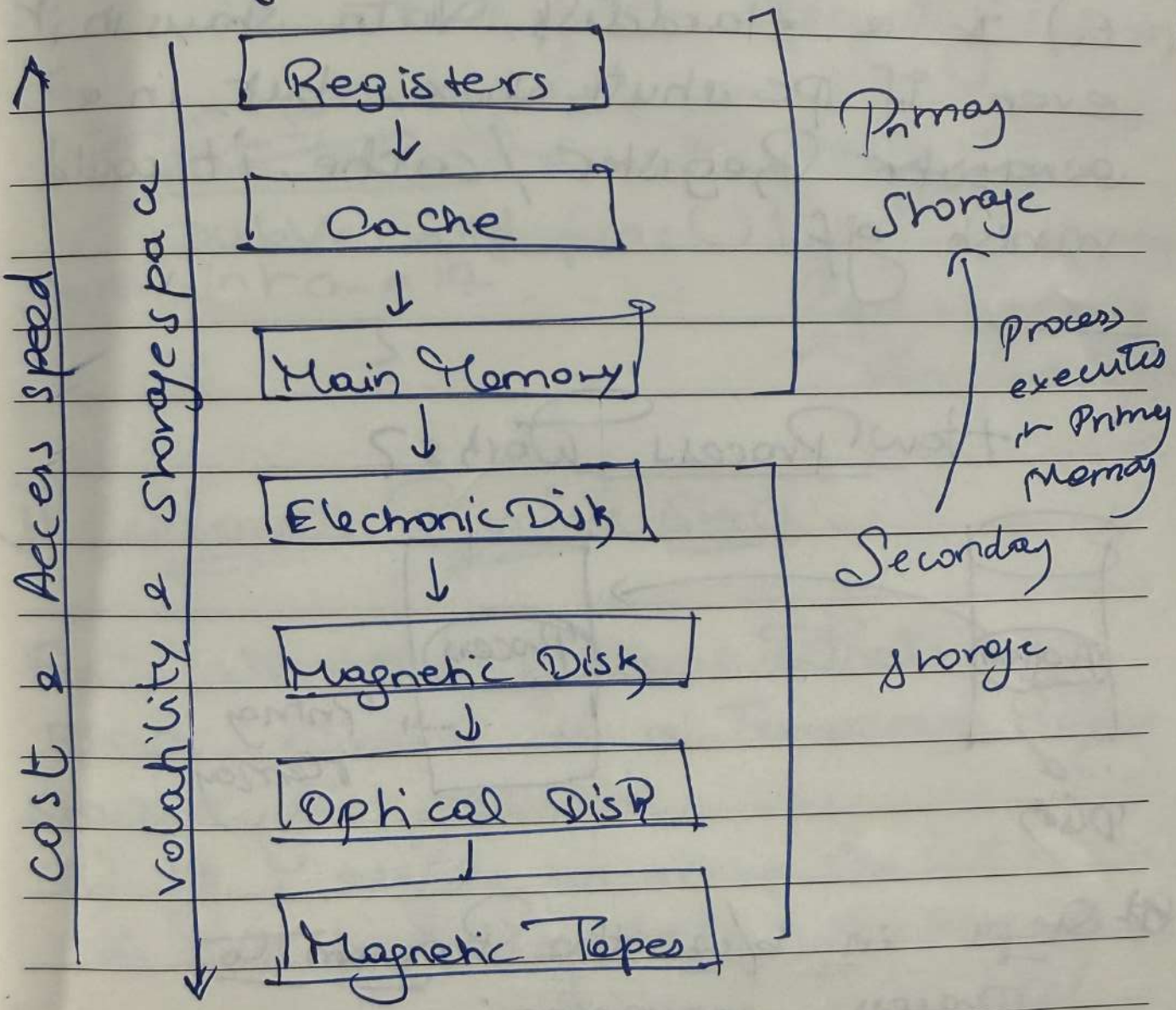
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Operating Systems

Types of storages



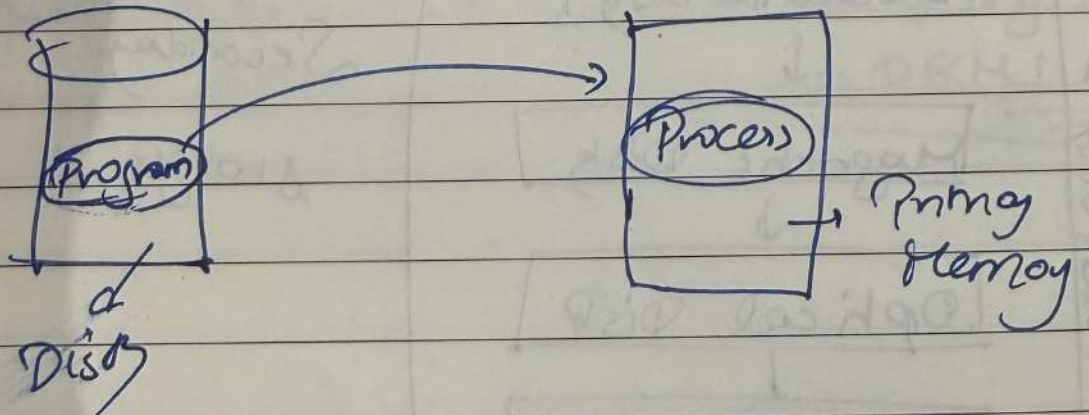
Registers are the costliest because of the high quality & millions of transistors present in it.

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Volatility refers to the data getting stored even after a shutdown,
Ex) In a Harddisk, data stays in it even if pc shuts down, but in a ~~computer~~ Register / cache, it could vanish off

How Process Works?



Steps in b/w the Program to Process conversion:

(i) Load program & static data to Memory.

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Static Data → it is the data used for initialization & stays till the end of the lifecycle of Program

Ex) present inside the function

~~int a = 10;~~ → Static, since it works whenever program runs & not only when func called

```
public void func(){
static int a = 10;
}
```

(ii) Allocate Runtime Stacks.

First, we need to know what's a stack. Stack is a Temporary Memory storage for functions.

Ex) If I define an array size is 5, it will be allocated in the Memory.

↓

Predefined Memory space.

↓

cleared once the job is done.

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* Therefore, when process runs, Stack will be Allocated.

(iii) Allocate Heap:

Heap is Dynamic Memory Allocation which is, when the user doesnot predefine memory space & it will be decided only in the Runtime, then it is Dynamic Alloc. Memory.

e.g) User i/p decides how much storage be occupied → Dynamic.

∴

Will be Allocated a space for Dynamic Memory Alloc.

(iv) I/O Tasks / operations

Three major tasks:

	File Descriptor
* I/p handle → Stdin →	0
* O/p handle → Stdout →	1
* Error handling → Stderr →	2

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(v) Handover of control to main:

Once OS transfers the process into memory, C Runtime will look for the main function & start of the Program. Now, the control is given to the main func. to execute & produce o/p. If, it sees:

return 0 → Successful execution

return 1/ -1 → Error, unsuccessful execution.

